

Functional Keywords

Unlike access modifiers, **Non-Access Modifiers** do not control visibility. Instead, they provide specialized properties to classes, variables, and methods.

IMMUTABILITY

1. The 'final' Keyword

Use `final` to freeze a value or prevent a class from being inherited.

```
final int PI = 3.14159; // Immutable variable
final class Secure { } // Cannot be extended
```

CLASS LEVEL

2. The 'static' Keyword

Attributes and methods marked `static` belong to the **class blueprint** itself, rather than individual objects.

```
static void greet() { System.out.println("Hello!"); }

// Called without creating an object instance:
Main.greet();
```

3. The 'abstract' Keyword

An abstract class cannot be instantiated. An abstract method has no body and **must** be overridden by a subclass.

```
abstract class Animal {  
    public abstract void makeSound(); // No body here  
}
```

Summary Table

Modifier	Applied To	Primary Effect
final	Class, Method, Variable	Prevents modification/inheritance.
static	Method, Variable	Shared by all instances of the class.
abstract	Class, Method	Enforces implementation in subclasses.
synchronized	Method	Thread-safety (one thread at a time).